UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

NATIONAL ENERGY TECHNOLOGY LABORATORY

Development of YIG Single-Crystal-Fiber Magnetic Field Sensor for Power-Grid Applications

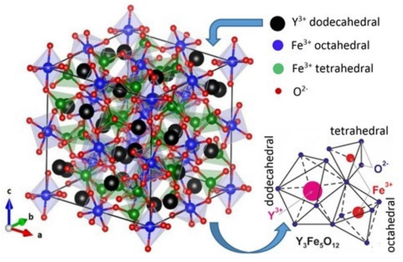
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INTRODUCTION

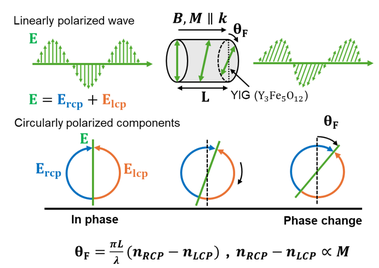
- Magnetic-field monitoring provides a non-contact diagnostics of power-grid assets and electromagnetic operating conditions.
  → It can support current-state estimation, fault detection, and equipment health monitoring
- Fiber-optic magnetic sensors are attractive for high-voltage and EMI-rich environments.
  → The optical sensing path enables electrical isolation, remote readout, and compatibility with harsh electromagnetic environments.
- Yttrium iron garnet (YIG, $Y_3Fe_5O_{12}$) is a ferrimagnetic magneto-optic crystal with strong Faraday rotation near telecom wavelengths.
  → YIG and related garnets are widely used in Faraday rotators, optical isolators, and magnetic field sensing components.
- LHPG enables YIG to be processed from a bulk-magneto-optic crystal into a single-crystal fiber geometry
  → This geometry may bridge bulk YIG Faraday elements and fiber-coupled magnetic-field sensors.

Goal: Demonstrate fiber-coupled, transmission-type magnetic-field sensing using an LHPG-grown YIG-SCF.

Unit cell of yttrium iron garnet ($Y_3Fe_5O_{12}$) [1]



Faraday Rotation Mechanism



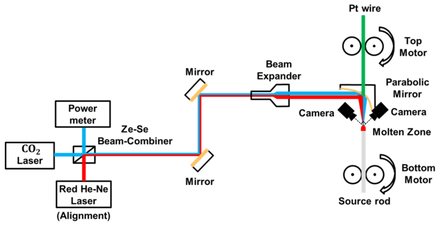
$$\theta_F = \frac{\pi L}{\lambda}(n_{RCP} - n_{LCP}) \ , \ n_{RCP} - n_{LCP} \propto M$$

[1] A. J. Princep et al., npj Quantum Mater., vol. 2, no. 1, p. 63, Nov. 2017, doi: 10.1038/s41535-017-0067-y.

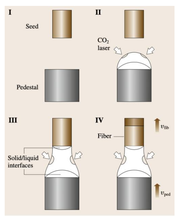
MATERIAL FABRICATION

- YIG single-crystal fibers (SCF) were grown by laser-heated pedestal growth (LHPG)
  → A focused CO_2 laser creates a localized molten zone between the YIG source rod and seed fiber.
- LHPG enables crucible-free growth of oxide single-crystal fibers.
  → This avoids crucible contamination with small-diameter fiber geometries.
- YIG is an incongruently melting oxide, which makes stable molten-zone growth challenging.
  → Phase stability depends strongly on temperature composition, pulling rate, and molten zone shape.
  → B_2O_3-assisted growth was investigated to improve processability of YIG-SCFs via LHPG.

Laser-Heated Pedestal Growth (LHPG) Setup [2]



Molten-Zone & Fiber Growth Process [3]

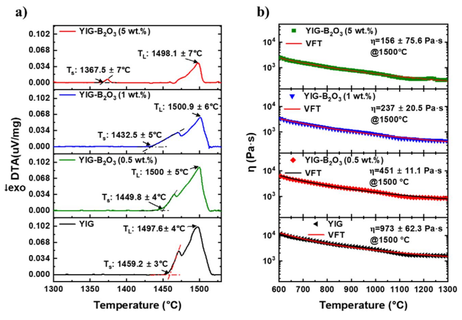


[2] Hong, J. Y.; Karki, D.; Sridar, S.; Ochodnicki, P. *Cryst. Growth Des.* 2026, *26*, 2473–2487. DOI: 10.1021/acs.cgd.5c01776.
[3] Andreeta, M. R. B.; Hernandes, A. C. "Laser-Heated Pedestal Growth of Oxide Fibers."; Springer, pp. 393–432. DOI: 10.1007/978-3-540-74761-1_13.

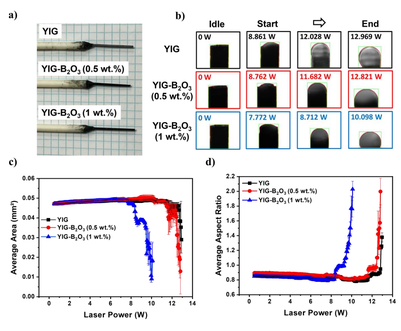
MATERIAL CHARACTERIZATION

- B_2O_3 addition lowers the solidus temperature (T_S) of YIG source pellets.
  → Lower T_S indicates that liquid-phase formation begins at a lower temperature, which facilitates molten-zone initiation.
- B_2O_3 reduces melt viscosity by 2~6x near the LHPG growth temperature (@1500 ˚C).
  → Viscosity (η) decreases from 973 ± 62 Pa · s for YIG to 451 ± 11 Pa · s, and 237 ± 21 Pa · s for 0.5, 1 wt.% B_2O_3, respectively.
  → Lower viscosity suggests improved melt deformability and molten-zone shape response.
- Molten-zone imaging defined the pre-growth process window for LHPG
  → Stable zone shape and controlled deformation are required before continuous fiber pulling

Thermal and Viscosity Behavior [2]

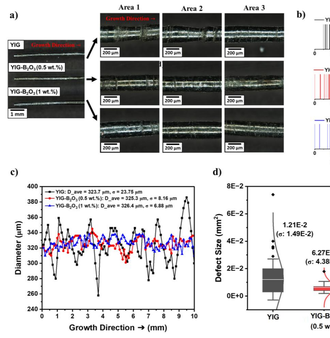


Molten-Zone Formation Window [2]

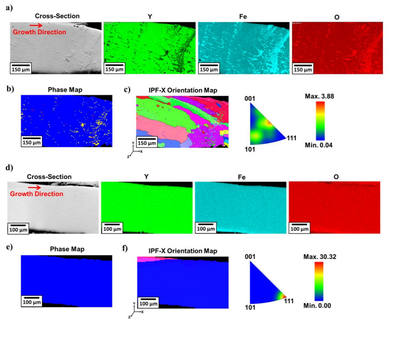


- B_2O_3-assisted YIG-SCFs showed more uniform diameter and fewer visible defects than undoped YIG.
  → The measured diameters remained in the hundreds-of-microns range, suitable for compact magneto-optic inserts.
- B_2O_3-assisted YIG-SCFs exhibited enhanced phase purity and more homogeneous Y-Fe-O distributions.
  → Undoped YIG contained secondary phases, including Fe_3O_4-rich regions, while YIG-B_2O_3 (1wt.%) was ~99.9% YIG phase.
- B_2O_3-assisted growth improved crystallographic orientation consistency near the [111] direction.
  → EBSD/IPF mapping showed ~ 10x higher crystallographic consistency [111] for the YIG-B_2O_3 (1wt.%) sample.

Diameter Uniformity and Defects Density [2]



Composition and Crystal Orientation [2]

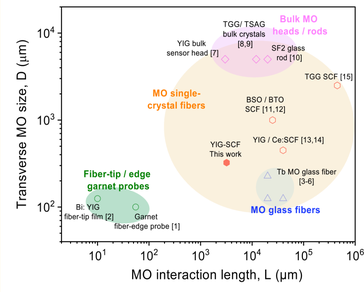


[3] Hong, J. Y.; Karki, D.; Sridar, S.; Ochodnicki, P. *Cryst. Growth Des.* **2026**, *26*, 2473–2487. DOI: 10.1021/acs.cgd.5c01776.

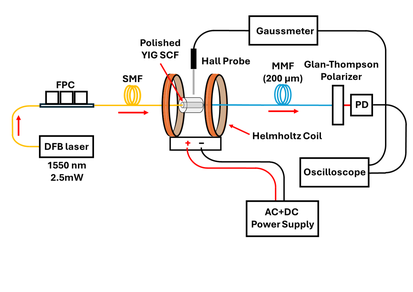
PRELIMINARY RESULTS & PERSPECTIVES

- YIG-SCF bridges bulk magneto-optic heads and fiber-form sensing elements.
  → The fiber geometry combines a sub-mm transverse dimension with a mm-cm magneto-optic interaction length.
- A transmission-type fiber-coupled setup was used for first-pass sensing tests.
  → 1550 nm light was launched through SMF, transmitted through polished YIG-SCF, collected by MMF, and analyzed with a Glan-Thompson polarizer.

Magneto-Optic Form-Factor Map

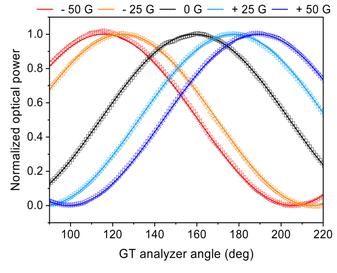


Fiber-Coupled Faraday Test Setup

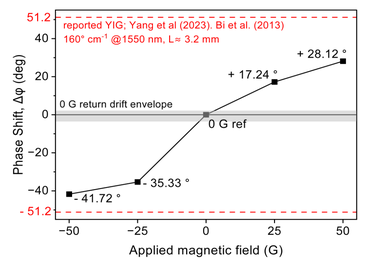


- Magnetic field produced a systematic shift in the Malus-type analyzer response.
  → Positive and negative fields shifted the response in opposite directions.
- The apparent phase shift increased with applied field magnitude.
  → Drift-corrected shifts were approximately +17.2° at +25 G, +28.1° at +50 G, −35.3° at −25 G, and −41.7° at −50 G.
- Current limitations are dominated by coupling, polarization drift, and thermal stability.
  → Future work will focus on improved packaging, calibrated field response, and AC magnetic-field sensing.

Field-Dependent Faraday Rotation Response



Phase Shift vs. Applied Magnetic Field



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